



8/7/2026

ASSIGNMENT DAY1

Introduction to Networking Hardware.

TASK 1.

- ❖ Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

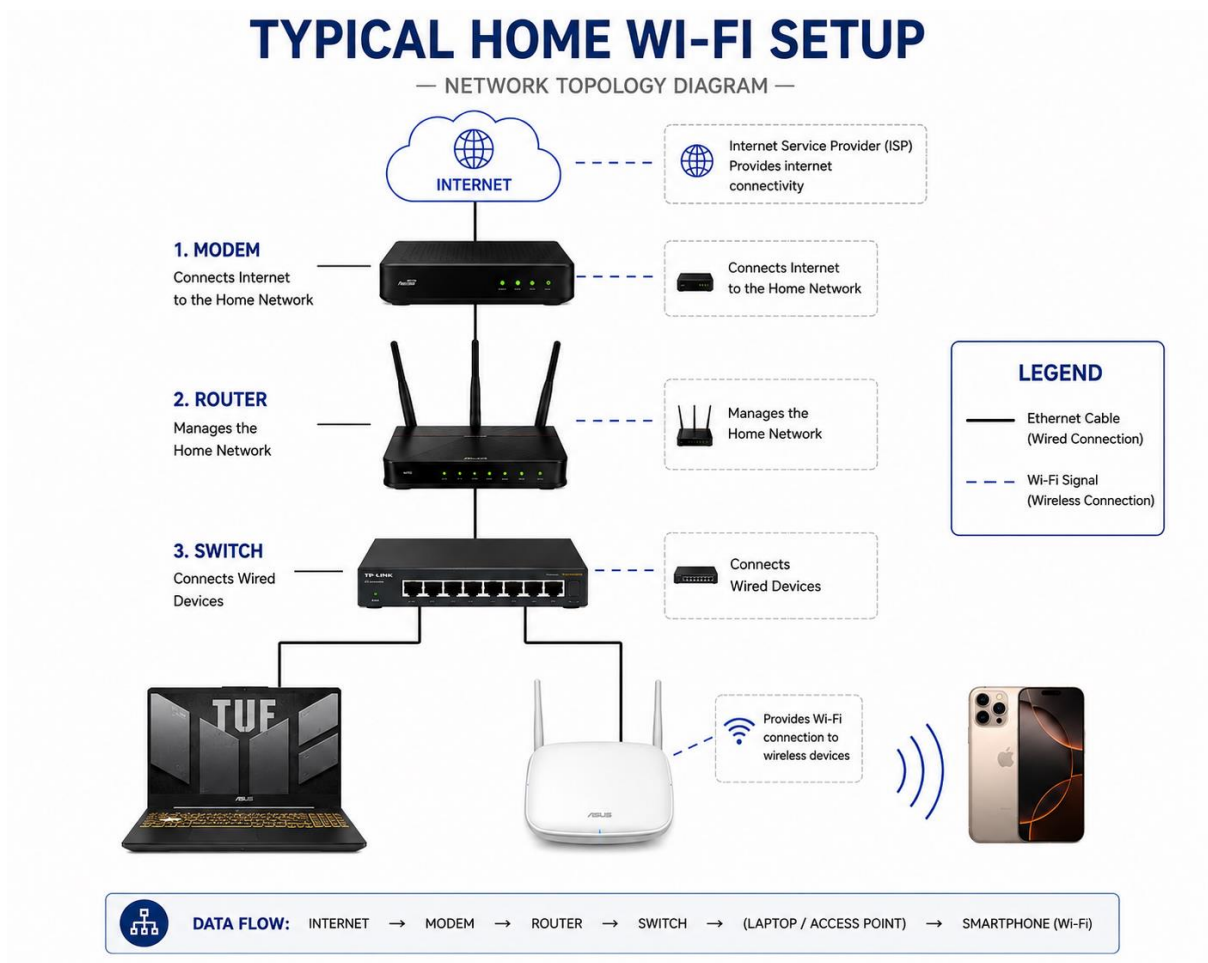


Figure : Typical Home Wi-Fi Network Topology.

Explanation :

- ✓ The This diagram illustrates a typical home Wi-Fi network. The modem connects the Internet to the router. The router connects to the switch

and access point. The laptop uses a wired connection, while the smartphone connects wirelessly through the access point.

TASK 2.

❖ **List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').**

➤ **Main Functions and Real Life Examples of Networking Hardware:**

<u>Networking Hardware</u>	<u>Main Function</u>	<u>Real Life Example</u>
Modem	Connects the home network to the internet.	Used at home to get an Internet connection from the ISP.
Router	Shares the internet and manages the home network.	Used at home to provide Wi-Fi to multiple devices.
Switch	Connects multiple wired devices in a network.	Used in offices to connect computers and printers.
Access Point	Provides Wi-Fi access to wireless devices.	Used in schools, hostels, and large homes to extend Wi-Fi coverage.
Network Cable	Transfers data through a wired connection.	Used to connect a computer, router, or switch.

TASK 3.

- ❖ Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.



Figure 1: Ethernet Cable.

- ✓ **Source:** Self generated (Created using Claude AI).
- ✓ **Common Use:** Used to connect computers, routers, and switches in home and office networks.



Figure 2: Coaxial Cable.

- ✓ **Source:** Self generated (Created using Claude AI).
- ✓ **Common Use:** Commonly used for cable TV and broadband Internet connections.

🌈 *Figure 3 (Continued on the Next Page).*



Figure 3: Fiber Optic Cable.

- ✓ **Source:** Self generated (Created using Claude AI).
- ✓ **Common Use:** Used for high speed Internet and long distance data communication.

TASK 4.

- ❖ Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

➤ Comparison Between Router and Switch :

<u>Feature</u>	<u>Router</u>	<u>Switch</u>
Main Use	Connects different networks and shares the Internet.	Connects multiple devices within the same network.
Typical Location	Used at home, offices, and schools.	Used in offices, schools, and computer labs.
Data Handling	Sends data between different networks using IP addresses.	Sends data to the correct device using MAC addresses.
Internet Connection	Connects the local network to the Internet.	Does not connect directly to the Internet.